

Sravani Kaviti

Email: sravanika76@gmail.com

Address: Dresden, Germany

LinkedIn: [linkedin.com/in/sravani-kaviti](https://www.linkedin.com/in/sravani-kaviti)

GitHub: github.com/sravanikaviti13

Portfolio: your-portfolio.com



Professional Summary

Machine Learning & Applied AI Engineer with 4+ years of experience developing and deploying real-time perception systems and machine-learning applications. Skilled in deep learning, model evaluation, and edge deployment using Python, PyTorch, ROS 2, ONNX, TensorRT, and NVIDIA Jetson.

Experience

Computer Vision Engineer (AV Safety Evaluation)

Jan 2026 – Present

FSD Fahrzeugsystemdaten GmbH

Dresden, Germany

- Developed and deployed a ROS 2 multi-camera perception pipeline on NVIDIA Jetson Orin NX, combining object detection, stereo depth, 3D tracking, traffic-sign recognition, and time-to-collision estimation for safety evaluation.
- Fine-tuned and benchmarked YOLOv8 and YOLO11 models for road-object detection, selecting YOLO11m for longer-range inference; achieved mAP50 of 0.68 on a road-scene subset and exported PyTorch weights through ONNX to TensorRT engines for Jetson deployment.
- Built a two-stage traffic-sign recognition pipeline using YOLO detection and EfficientNet-B0 classification, achieving 98.34% validation accuracy across 211 German traffic-sign classes.
- Integrated ZED SDK stereo depth and 3D object tracking with custom ID stabilization and TTC logic to calculate object distance, relative velocity, and time to collision; achieved 7–10 FPS during on-road testing.
- Designed a prototype ego-lane detection module by fusing a KITTI-trained ResNet-18 U-Net with UFLD lane-line detection, achieving IoU scores of 0.913 for road segmentation and 0.924 for ego-lane segmentation.

Applied AI Engineer

Jan 2025 – Dec 2025

Infineon Technologies

Dresden, Germany

- Developed a Workshop Management Application adopted by approximately 200 employees within three months, supporting workshop creation, moderator assignment, feedback, and administrative workflows.
- Built and evaluated a LambdaMART moderator-recommendation system using skill, location, and feedback features, achieving NDCG@5 of 0.68 on 20 held-out workshops and a 25% improvement over a skill-and-location baseline.
- Implemented moderator-management, approval, and notification workflows, supporting approximately 15 workshops during the initial three-month rollout.
- Implemented LLM-based semantic matching to evaluate participant responses in a gamified learning platform using generative AI, reaching more than 100 active learners among 500+ employees.

Data Analyst

Jan 2024 – Dec 2024

Infineon Technologies

Dresden, Germany

- Integrated over 80K records from planning and equipment databases using SQL, reconstructing equipment-change histories and linking gas, energy, and dose parameters to setup times.
- Performed exploratory, statistical, and clustering analysis to identify equipment and process-parameter combinations associated with setup-time variation and longer setup durations.
- Trained Random Forest, XGBoost, and Gradient Boosting models in KNIME for setup-time prediction, integrating SQL queries and Python scripts; selected XGBoost with an R-squared of 0.57.
- Collaborated with domain teams to identify additional setup-time drivers and document cross-application data requirements for future model improvement.

Software Engineer (Backend)

Cognizant Technology Solutions

Aug 2021 – Aug 2023

Hyderabad, India

- Implemented Oracle Forms and SQL and PL/SQL solutions for an insurance claims platform, delivering three new forms from requirements through testing and supporting production deployment via Jenkins-based CI/CD workflows.
- Enhanced a Microsoft VBA documentation tool by improving macro performance and security, while automating a frequently generated report and reducing manual refinement and review effort by approximately 40%.
- Investigated data and production issues through SQL, ServiceNow, and stakeholder coordination, handling approximately 25 tickets weekly and generating census files for the insurance platform.

Skills

Programming and Data: Python, C/C++, SQL, PL/SQL, NumPy, Pandas, Matplotlib

Machine Learning: PyTorch, scikit-learn, Transformer models, XGBoost, LambdaMART, clustering, SHAP

Computer Vision: YOLO, OpenCV, EfficientNet, U-Net, object detection, multi-object tracking, semantic segmentation, camera calibration, transfer learning

Generative AI and NLP: Conditional diffusion models, RAG, vector databases, LLM-based semantic matching

Robotics and Edge Deployment: ROS 2, sensor synchronisation, GPS/IMU, ZED SDK, ONNX, TensorRT, CUDA, NVIDIA Jetson, Docker

Tools: KNIME, Git, GitHub, Jenkins, ServiceNow, Linux/Unix

Languages: English (C1), German (B1)

Education

Master of Science in Visual Computing

2023 – 2027

Technische Universität Dresden

Dresden, Germany

- **Thesis:** Machine-Interpretable Traffic Regulations for Autonomous-Driving Compliance
- **Relevant Coursework:** Machine Learning, Computer Vision, Deep Learning for Vision, Data Management, Data Visualization, Touch Sensing, Deep Neural Network Hardware and Large Language Models.

Bachelor of Technology in Electronics and Communications Engineering

2017 – 2021

V.R. Siddhartha Engineering College

Vijayawada, India

- **Grade:** 1.9
- **Thesis:** Mobile Detacher Using a Heart Rate Monitoring System
- **Relevant Coursework:** Data Structures & Algorithms, Computer Networks, Embedded C, MATLAB, Microcontroller Programming, Digital Signal Processing.

Projects

Conditional Diffusion for Tactile Data Augmentation

- Collected and processed approximately 12K DIGIT tactile images from a UFactory xArm7, pairing images with 19-dimensional kinematic deltas covering joint angles, end-effector pose, and force–torque measurements.
- Developed and trained a conditional diffusion model using a U-Net with DINOv2 image features and kinematic-delta cross-attention to generate tactile images directly from changes in robot state.
- Evaluated generated tactile images across single- and multi-object variants using IoU, MSE, PSNR, and SSIM, achieving 89.2% usable reconstructions on held-out pairs and 68.6% on unseen spatial gel regions.

Explainable Multi-Label Emotion Detection

- Fine-tuned RoBERTa and ALBERT transformer models for multi-label emotion classification using PyTorch and Hugging Face Transformers, evaluating 13 configurations and achieving a mean F1 of 0.782 with RoBERTa.
- Applied SHAP token attribution and human evaluation to assess model explanations and robustness, observing a mean F1 of approximately 0.52 on manually labeled samples.